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Portfolio: <https://dgrOne.github.io/dataportfolio>

AI & Data Specialist | Industrial Automation & Analytics Specialist

PROFILE SUMMARY

AI and Data professional with 7+ years of hands-on experience in manufacturing and energy operations, now specializing in AI-driven automation and data intelligence. Strong at translating real-world operational challenges into scalable digital solutions using Python, Power BI, SQL, and modern AI frameworks.

Brings deep domain knowledge in maintenance, utilities, and SAP-driven operations, combined with practical experience building AI-powered workflows, dashboards, and automation systems. Experienced in applying Generative AI, RAG-based systems, and agentic workflows to improve decision-making, reduce manual effort, and enable data-driven operations.

Known for strong execution, adaptability, and the ability to bridge the gap between engineering teams and business stakeholders.

CORE SKILLS

AI, Automation & Engineering

- Generative AI (LLMs, Prompt Engineering, Agentic Workflows)
- **Agentic AI:** Design and deployment of autonomous agents (Gemini, OpenAI, LangGraph, CrewAI, BeeAI, Autogen).
- RAG (Retrieval-Augmented Generation)
- LangChain, LangGraph
- OpenAI API, Model Context Protocol
- AI Orchestration & Workflow Design
- AI-based Automation (n8n, App Script)
- Responsible AI & Prompt Optimization
- Gemini API Integration
- **Cloud & Integration:** Google Cloud Platform (GCP), Serverless Automation (Apps Script Triggers), API Orchestration, Webhooks.

Data & Analytics

- Python (Pandas, NumPy, Matplotlib, Seaborn)
- SQL (Joins, Aggregations, Data Modeling)
- Power BI (DAX, Power Query, Data Modeling)
- Advanced Excel (VBA, Macros, Power Query)

- Data Analysis, Reporting, Visualization

Platforms & Tools

- SAP HANA (Maintenance & Procurement)
- Supabase (SQL Backend)
- VS Code, Jupyter Notebook
- Microsoft Power Platform
- API Integration
- Hugging face
- Github
- LLMs (Chatgpt, Copilot, Claude, Perplexity, Gemini)

PROJECT EXPERIENCE

AI-Powered Procurement Expediting Agent Tools: Google Cloud (Apps Script), Gemini 1.5 Flash API, Telegram Bot API, JavaScript

- Engineered an end-to-end agentic workflow to automate supplier follow-ups for overdue Purchase Orders.
- Integrated **Gemini 1.5 Flash** to generate context-aware, multi-level escalation messages (polite to firm) based on the severity of delivery delays.
- Automated real-time tracking and WhatsApp-ready nudges via Telegram, reducing manual administrative overhead by **~50%**. (Human in the loop)
- Implemented automated sheet logging to track follow-up history and escalation levels without manual data entry.
- Developed a **serverless cloud automation** that runs autonomously via time-driven triggers to monitor global procurement data.
- **Time Savings:** Reduced manual follow-up time by **~10 hours/week**.

AI-Driven PM Scheduling Automation

Tools: Excel, VBA, ChatGPT

- Automated preventive maintenance scheduling from yearly master data
- Reduced manual effort by **12+ hours/month** across 400+ assets
- Improved scheduling accuracy and compliance tracking

Technical Maintenance Support Agent Tools: Python, Hugging Face, RAG (Retrieval-Augmented Generation), LangChain, Langgraph

- Developed and deployed a specialized AI agent on **Hugging Face** to provide instant technical support for maintenance teams.
- Utilized **RAG** to index complex machinery manuals and SOPs, Breakdown history, allowing engineers to query repair procedures using natural language.

- Reduced "Time-to-Information" during critical breakdowns, directly supporting a reduction in **MTTR (Mean Time to Repair)**.
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Procurement Specification Verification Bot Tools: Google sheets/Excel, Google Cloud (AppScript), Telegram/Gemini API, NLP, Gemini 3 Flash

- Designed a Telegram-based bot to assist the procurement team in instantly verifying spare part specifications.
 - Bridged the communication gap between technical engineering requirements and purchasing teams by providing real-time spec validation.
 - Improved the accuracy of spare part ordering, reducing "wrong-part" procurement errors.
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AI-Powered Inventory Management System

Tools: Supabase, SQL, Bolt AI

- Designed a web-based inventory tracking system(webapp)
 - Enabled real-time stock visibility and consumption tracking
 - Improved audit transparency and procurement planning
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AMC Renewal Automation System

Tools: Google Cloud (App Script), JavaScript

- Automated renewal reminders with email alerts
 - Reduced missed renewals and manual follow-ups
 - Improved compliance and vendor coordination
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PROFESSIONAL EXPERIENCE

Nordex Acciona Windpower Pvt. Ltd

Operational BI & AI Automation Specialist | Chennai | Jun 2022 – Present

- **Architected AI-Driven Procurement Agents:** Engineered autonomous workflows using **Gemini 1.5 Flash** and **Google Cloud** to monitor SAP PO/PR status and execute multi-level supplier follow-ups, reducing manual oversight by **50%**.
- **Automated Inventory & Procurement Planning:** Developed advanced **VBA-powered models** for automated stock analysis and inventory forecasting, ensuring real-time visibility and optimized spare parts availability.
- **Streamlined Executive Reporting Pipelines:** Orchestrated cross-departmental data collection to automate monthly **Executive Management Presentations**, significantly reducing the lead time and manual effort for performance reporting.

- **Engineered Serverless Cloud Automations:** Built a cloud-based AMC Coordination system via **Google Apps Script** to manage renewals for 100+ critical assets, eliminating compliance risks and manual tracking.
 - **Optimized Maintenance & Utility Operations:** Automated preventive maintenance (PM) scheduling for 400+ assets and utility tracking (Power/Water/Fuel) using **Advanced Excel & Macros**, saving **12+ man-hours/month**.
 - **AI Implementation Strategy:** Successfully integrated Generative AI into daily workflows, automating **20–30%** of recurring administrative tasks through custom scripts and intelligent prompt engineering.
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MRF Ltd – Electrical Supervisor

Aug 2021 – Apr 2022

- Analyzed SAP data for transformer and DG set performance
 - Supported preventive maintenance through structured data logging
 - Improved data accuracy for long-term asset analysis
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TI Metal Forming – Diploma Trainee

Mar 2019 – Aug 2021

- Maintained power and maintenance logs
 - Supported data collection for operational efficiency tracking
 - Built strong foundation in industrial systems and data accuracy
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EDUCATION & CERTIFICATIONS

Diploma – Electrical & Electronics Engineering

Murugappa Polytechnic College | 2018

Certifications

- Advanced Data Visualization – Power BI (GUVI – HCL)
 - Generative AI & Prompt Engineering (UPTOR)
 - Data Analytics Specialization – Novitech
 - Machine Learning Internship – Novitech
 - IBM RAG Agentic AI (IBM)
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LANGUAGES

- Tamil – Fluent
- English – Fluent
- Telugu – Conversational

